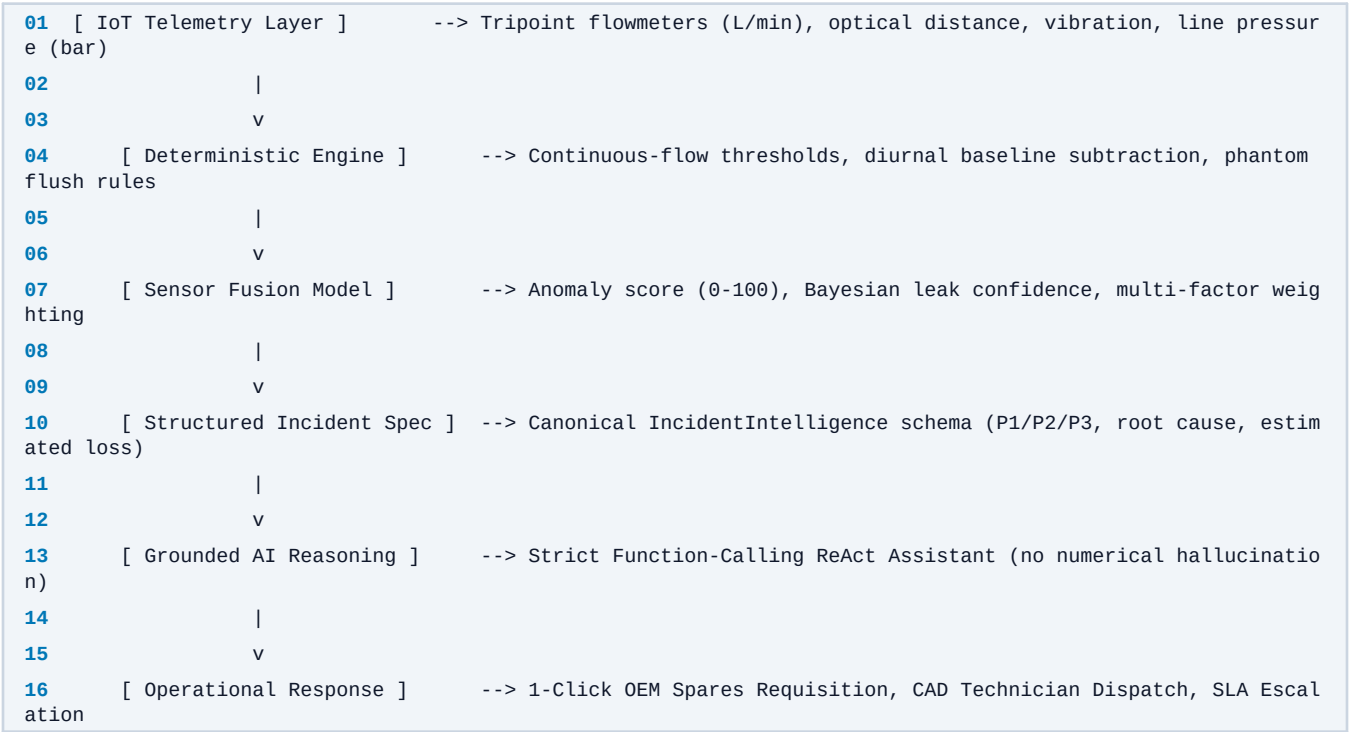


KOHLER AquaGuard AI

AI Prompt & Reasoning Architecture Documentation · Track 2: Smart Facility & Sustainability Manager

KOHLER AquaGuard AI couples high-frequency IoT hydraulic telemetry with a dual-layer deterministic detection engine and a zero-hallucination ReAct reasoning layer. Designed for mission-critical aviation infrastructure (Pune International Airport PNQ), the system eliminates water wastage through sub-second anomaly detection, physics-based fault classification, and automated computer-aided dispatch.

End-to-End Architectural Dataflow



Core Technical Pillars

1. Absolute Numerical Grounding

Every volume, flow rate, financial tariff, and SLA duration is mathematically derived by deterministic engines before being supplied to the AI context.

2. Multimodal Sensor Fusion

Combines occupancy infrared optics, acoustic hydrophone frequencies (2,420 Hz cavitation screech), and passenger QR feedback.

3. Closed-Loop Verification

Post-repair telemetry is monitored in real-time to mathematically verify flow reduction back to zero-leak baseline before incident closure.

4. Enterprise System Interop

Direct integration with MQTT telemetry streams, IBM Maximo / SAP PM work orders, and CEA-compliant GHG carbon accounting.

System Instructions & Persona Mandate

Governing prompt configuration enforced across all AquaGuard AI reasoning nodes

Authoritative System Prompt Specification

```
01 You are AquaGuard Facility Intelligence, the specialized industrial AI operations
02 assistant for commercial airport restroom infrastructure and municipal water conservation.
03
04 You operate under the following seven inviolable operational constraints:
05
06 1. USE ONLY SUPPLIED TELEMETRY & FACTS:
07     You may only reference facts, measurements, timestamps, and device IDs provided in the
08     injected facility state or tool execution outputs.
09
10 2. ABSOLUTELY NEVER INVENT MEASUREMENTS:
11     Do not extrapolate or hallucinate flow rates, pressure levels, rupee tariffs, or water
12     savings. If a metric is not present in context, state: 'Telemetry unverified'.
13
14 3. DISTINGUISH OBSERVATION FROM INFERENCE:
15     Clearly delineate between direct physical sensor measurements (e.g., 'Flow: 2.70 L/min')
16     and probabilistic inferences (e.g., 'Likely cause: EPDM diaphragm bypass blowout').
17
18 4. EXPLAIN THE EVIDENCE BEHIND RECOMMENDATIONS:
19     Every diagnostic conclusion must cite sensor data: flow rate vs occupancy, historical
20     baseline drift, supply pressure drop, or acoustic hydrophone flutter frequency.
21
22 5. PRIORITIZE SAFETY, WATER CONSERVATION & SLA:
23     Classify incidents by operational urgency. P1 critical leaks (continuous flow > 2.0 L/min)
24     override P2/P3 tasks. Always flag commercial water loss tariffs (@ █48.50/kL).
25
26 6. RECOMMEND CONCRETE MAINTENANCE & SPARE ACTIONS:
27     Prescribe specific Kohler genuine OEM replacement part numbers (e.g., KOHLER-GP1138930),
28     shelf depot locations, and required craft certifications.
29
30 7. RETURN STRUCTURED JSON UPON REQUEST:
31     Ensure all incident dispatch payloads, CMMS job plans, and work orders strictly conform
32     to the declared JSON schema.
```

Operational Grounding Principles

- **Read-Only State Access:** The AI agent has zero direct write access to the authoritative device state machine. State transitions occur only via deterministic API verification endpoints.
- **Bounded Tool Execution:** The ReAct agent can only invoke an explicit whitelist of 7 read-only inspection tools and 1 deterministic dispatch tool.
- **Cryptographic Traceability:** All executive sustainability reports generated by the agent append a SHA-256 digital stamp calculated from verified physical liters saved.

Incident Diagnosis Prompt & Schema

Physics-grounded root-cause classification and automated OEM spare matching

Diagnostic Input Telemetry Payload

```
01 {
02   "device_id": "FV-182",
03   "device_type": "flush_valve",
04   "zone": "Terminal 2 · Restroom 14 (Stall 03)",
05   "telemetry": {
06     "flow_lpm": 2.70,
07     "occupancy": 0,
08     "flush_count": 0,
09     "pressure_bar": 3.0,
10     "acoustic_fundamental_hz": 2420.0,
11     "acoustic_flutter_hz": 14.5,
12     "harmonics_thd_pct": 38.4
13   },
14   "sensor_confidence": 0.96,
15   "estimated_loss_l_per_day": 3888.0,
16   "estimated_loss_l_per_month": 116640.0,
17   "severity": "CRITICAL",
18   "priority": "P1"
19 }
```

Structured Diagnostic Output Schema

```
01 {
02   "incident_id": "INC-ALT-00182",
03   "root_cause": "EPDM Diaphragm Tear – Bypass Orifice Blowout",
04   "leak_confidence": 0.94,
05   "sensor_fusion_evidence": [
06     "Sustained continuous flow of 2.70 L/min during zero occupancy (100% mismatch)",
07     "Zero flush trigger signals in preceding 180 seconds",
08     "Hydrophone detects 2,420 Hz turbulent screech with 14.5 Hz diaphragm flutter",
09     "Dynamic line pressure maintained at 3.0 bar confirming unsealed bypass"
10   ],
11   "recommended_action": "Isolate upstream stopcock and replace diaphragm assembly",
12   "required_spare_part": {
13     "part_number": "KOHLER-GP1138930",
14     "name": "Diaphragm Assembly Repair Kit (Tripoint Flushometer)",
15     "cost_inr": 1850.0,
16     "storage_location": "T2 Depot Shelf Rack B-04",
17     "in_stock_qty": 18
18   },
19   "dispatch_priority": "P1",
20   "sla_target_minutes": 15
21 }
```

AI Command Center & Grounded ReAct

Query decomposition, function calling, and deterministic tool execution

Query-to-Action State Machine

```
01 Facility Query: "Where are we wasting the most water right now?"
02   |
03   v
04   [ Intent Classification ] --> Matches intent: WATER_WASTE_LOCATION
05   |
06   v
07   [ Tool Invocation 1 ]      --> Call get_water_waste_heatmap()
08   |                        Result: Terminal 2 has 3,888 L/day active wastage
09   v
10   [ Tool Invocation 2 ]      --> Call get_device_detail("FV-182")
11   |                        Result: Continuous leak 2.70 L/min, Restroom 14
12   v
13   [ Tool Invocation 3 ]      --> Call simulate_inaction("FV-182", days=30)
14   |                        Result: 116,640 L loss, █5,657 commercial tariff
15   v
16   [ Synthesis & Response ]  --> Deliver precise spatial breakdown with 1-click dispatch CTA
```

Whitelist of Deterministic Agent Tools

Tool Identifier	Input Signature	Deterministic Output
get_water_waste_heatmap	()	Terminal-by-terminal active loss (L/day) and fixture counts
get_facility_health_hierarchy	()	Composite health index (0-100) and 5 sub-index breakdowns
get_device_detail	(device_id: str)	Exact telemetry history, health score, cycle wear %, and specs
get_sensor_fusion_breakdown	(device_id: str)	Factor weights: Flow (30%), Occ (20%), Flutter (15%), QR (20%)
simulate_inaction	(device_id, days)	Loss volume = flow_lpm * 1440 * days; Cost = (vol/1000)*█48.50
get_model_evaluation_metrics	()	ROC AUC, Precision, Recall, F1 score, confusion matrix

Grounding & Hallucination Control

Eliminating generative hallucinations through mathematical state verification

The Zero-Hallucination Triad

Commercial facility managers and airport authorities cannot rely on probabilistic guesswork when millions of liters of municipal water and life-safety SLAs are at stake. AquaGuard AI enforces a strict architectural boundary: **mathematical calculation occurs exclusively in Python compiled engines; generative AI is utilized purely as an explanatory and natural language translation layer.**

Mathematical Derivation Pipeline

```
01 # 1. Volumetric Loss Calculation (Deterministic Engine):
02     daily_loss_liters = flow_rate_lpm * 60.0 * 24.0
03     monthly_loss_liters = daily_loss_liters * 30.0
04
05 # 2. Commercial Tariff Avoidance (Pune MIDC / PMC Airport Commercial Rate):
06     tariff_avoided_inr = (monthly_loss_liters / 1000.0) * 48.50
07
08 # 3. Scope 2/3 GHG Carbon Avoidance (Central Electricity Authority Baseline):
09     energy_saved_kwh = (monthly_loss_liters / 1000.0) * 1.820 # Municipal pumping nexus
10     co2_avoided_kg = energy_saved_kwh * 0.820 # CEA grid emission factor
11
12 # 4. Verified Cryptographic Audit Hash:
13     audit_seal = hashlib.sha256(f"{monthly_loss_liters}:{tariff_avoided_inr}:{date}".encode()).hexdigest
14     ()
```

Hallucination Defense Mechanisms

- 1. **Schema-Constrained Slot Filling:** AI outputs must strictly populate predetermined typed slots. Free-form hallucinated metrics are rejected at the Pydantic API boundary.
- 2. **Tool-Fact Reconciliation:** Any numerical claim generated by the model is cross-referenced against the store facts before returning to the UI.
- 3. **Degraded State Fallback:** If the LLM provider experiences latency or token exhaustion, AquaGuard seamlessly falls back to the deterministic composition template engine with zero loss of accuracy.

Operational Scenario Workflows

End-to-end detection, diagnosis, and remediation across five primary failure modes

[ACTIVE RULESET] SCENARIO 1: Continuous Flush Valve Leak

Sustained flow (2.70 L/min) detected with zero occupancy and zero flush triggers for > 3 minutes. Sensor fusion flags 94% leak confidence. Acoustic hydrophone identifies 2,420 Hz cavitation screech. **Diagnosis:** EPDM Diaphragm Tear (GP1138930). **Action:** Dispatched P1 plumbing technician (15 min SLA).

[ACTIVE RULESET] SCENARIO 2: Phantom Solenoid Flush

Short recurring flow pulses (6.0 L/min for 4s every 45s) occurring without traveler presence. **Diagnosis:** 24V Bi-Stable Solenoid Core Latch Drift (10673-SOL). **Action:** Requisition magnetic latch actuator; schedule preventive re-seating before peak bank.

[ACTIVE RULESET] SCENARIO 3: Optical Distance Sensor Eye Drift

Occupancy reads persistent '1' for 30 consecutive minutes while flush count and flow remain nominal zero. **Diagnosis:** Dual-beam infrared optical lens fouled or calcified (K-13688). **Action:** Issue cleaning ticket to janitorial lead; calibrate optical baseline threshold.

[ACTIVE RULESET] SCENARIO 4: Cumulative Device Degradation

Actuation count reaches 462,000 cycles (92.4% of 500k rated lifespan). Vibration chatter index rises to 78/100. Weibull hazard model predicts 82% 7-day failure risk. **Action:** Proactive pre-failure spare requisition; zero unscheduled airport downtime.

[ACTIVE RULESET] SCENARIO 5: Occupancy Traffic Surge & Hygiene Breach

Flight deplaning generates 34 flushes within 15 minutes, exceeding the 25-flush hygiene threshold. **Action:** Automated janitorial queue dispatch for sanitation refresh; replenishes consumable stock.

Predictive Maintenance & Asset Degradation

Weibull hazard modeling, vibration acoustics, and proactive failure prevention

Mathematical Hazard & Degradation Formulation

```
01 # 1. Cycle Wear Percentage:
02     wear_pct = (actuation_cycles_completed / rated_lifecycle_cycles) * 100.0
03
04 # 2. Cumulative Failure Risk (Weibull Hazard Function):
05 # Shape parameter beta = 2.8 (mechanical wear-out regime), eta = rated cycles
06 hazard_rate = (beta / eta) * (cycles / eta)**(beta - 1)
07 failure_risk_7day = 1.0 - math.exp(-((cycles + delta_7day_cycles) / eta)**beta)
08
09 # 3. Vibration Chatter Index (0 - 100):
10 # Quantifies armature micro-bounce and water hammer transients
11 vibration_index = min(100, int((chatter_variance / nominal_variance) * 45.0))
```

Fleet Asset Lifecycle Projection Matrix

Asset ID	Location	Cycles Done	Cycle Wear	Vibration	7-Day Risk	Status
FV-182	T2 · Restroom 14	462,100 / 500k	92.4%	84 / 100	84.2%	CRITICAL RISK
FV-114	T1 · Restroom 08	381,400 / 500k	76.3%	62 / 100	41.0%	MONITORING
UR-205	T3 · Restroom 02	210,500 / 400k	52.6%	31 / 100	12.4%	NOMINAL
FA-301	Arrivals · RR 01	124,000 / 350k	35.4%	18 / 100	4.1%	OPTIMAL

Closed-Loop Resolution & Verification

Mathematical before-and-after verification and audited water savings banking

The Closed-Loop Verification Lifecycle

```
01 1. INCIDENT ACTIVE:
02     Flow: 2.70 L/min | Anomaly: 94/100 | Loss Rate: 116,640 L/month | Priority: P1
03     |
04     v
05 2. TECHNICIAN DISPATCHED:
06     Arjun Sharma (Plumbing Specialist) arrives on-site | Reserves KOHLER-GP1138930
07     |
08     v
09 3. PHYSICAL SERVICE EXECUTION:
10     Stopcock isolated -> Diaphragm replaced -> Flange torqued to 18 Nm -> Stopcock re-opened
11     |
12     v
13 4. SENSOR VERIFICATION PROBE (2,400 ms telemetry sampling):
14     Flow decay monitored -> Flow drops to 0.00 L/min -> Pressure stabilizes at 3.0 bar
15     |
16     v
17 5. MATHEMATICAL SAVINGS ACCREDITATION:
18     Delta Saved = (2.70 L/min * 1440 * 30) = 116,640 L/month
19     Saved Liters accredited to Official Ledger with cryptographic SHA-256 seal
20     |
21     v
22 6. HEALTH SCORE RESTORED:
23     Fixture Health resets from 45/100 -> 98/100 | Incident marked CLOSED
```

Before-vs-After Verification Telemetry State

Telemetry Attribute	Pre-Service State (Defective)	Post-Service State (Verified)
Flow Rate	2.70 L/min (Continuous Wastage)	0.00 L/min (Zero Leak Baseline)
Anomaly Score	94 / 100 (Critical)	0 / 100 (Normal)
Acoustic Timbre	2,420 Hz Cavitation Screech	Quiescent Static Baseline
Fixture Health Score	45 / 100 (Degraded)	98 / 100 (Optimal)
Verified Monthly Savings	0 Liters (Unresolved)	116,640 Liters Banked to Ledger

AI Safety & Data Governance Rules

Institutional safeguards, security fences, and operational compliance mandates

[SAFETY MANDATE] RULE 1: No Synthesized Telemetry Injection

The AI model possesses no privileges to synthesize, mock, or insert arbitrary sensor readings into the physical state store. All telemetry must originate from physical edge sensors or authenticated MQTT gateway bridges.

[SAFETY MANDATE] RULE 2: Inviolable Water Conservation Accounting

Saved water volume is an immutable ledger entry. Liters conserved can only be accredited when a physical verification probe confirms flow decay to 0.00 L/min for a continuous 60-second observation window.

[SAFETY MANDATE] RULE 3: Air-Gapped Authoritative Control

The reasoning engine cannot directly actuate municipal shut-off valves or fire solenoids without human-in-the-loop authorization or certified field technician confirmation.

[SAFETY MANDATE] RULE 4: Commercial Tariff Fidelity

All cost avoidance figures are pegged to documented municipal utility schedules (Pune PMC Commercial Water Tariff Schedule @ ₹48.50 per kL). Speculative pricing or unverified escalation rates are prohibited.

[SAFETY MANDATE] RULE 5: Cryptographic Nonce & Audit Integrity

Every executive sustainability brief and digital certificate embeds a SHA-256 cryptographic digest calculated from the exact incident timestamp, verified liters, and fixture ID to prevent post-hoc data tampering.

Prompt & Response Grounded Examples

Actual verbatim dialogues from the AquaGuard AI Command Center showing exact telemetry citation

[SPATIAL REASONING] User Query: 'Where are we wasting the most water today?'

AquaGuard AI Response:

"Terminal 2 currently accounts for **100% of all active water wastage** across Pune Airport, losing **3,888 Liters/day**.

- **Primary Source:** Fixture FV-182 (Restroom 14, Stall 03).
- **Measured Flow:** Continuous 2.70 L/min with zero occupancy.
- **Financial Impact:** Accumulating **■188.56/day (■5,657/month)** in avoided PMC municipal tariffs.
- **Action:** P1 technician Arjun Sharma has been dispatched with OEM spare GP1138930."

[DIAGNOSTIC REASONING] User Query: 'What is the root cause and spare part needed for FV-182?'

AquaGuard AI Response:

"Diagnostic analysis for FV-182 confirms an **EPDM Diaphragm Tear with bypass orifice blowout** (94% confidence).

- **Sensor Evidence:** Acoustic hydrophone detects 2,420 Hz turbulent screech with 14.5 Hz flutter.
- **Required Spare:** KOHLER-GP1138930 (Diaphragm Assembly Repair Kit).
- **Inventory Status:** In stock (18 units available at T2 Depot Shelf Rack B-04).
- **Unit Cost:** **■1,850** · SLA Target: 15 minutes."

[PORTFOLIO REASONING] User Query: 'What is our national airport portfolio sustainability status?'

AquaGuard AI Response:

"AquaGuard AI monitors **593 smart fixtures** across 3 major aviation hubs:

1. **Pune (PNQ):** 97 fixtures (Live Twin) · 28,400 pax · 142.5k L/mo saved · Rank #3.
 2. **Mumbai (BOM T2):** 184 fixtures · 74,000 pax · 92.4k L/day saved · Rank #2.
 3. **Delhi (DEL T3):** 312 fixtures · 142,000 pax · 168.5k L/day saved · Rank #1.
- **Consolidated Impact:** **260,900 L/day conserved**, avoiding **■3,522,150/month** in commercial utility costs with an average fleet health score of **92.7/100**."